

CrossCov-U: Cross-Covariance Bases for Score-Faithful Sparse Attention

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Abstract

Sparse attention methods such as Loki cut the cost of decoding by ranking the tokens in the KV-cache with attention scores computed in a low-dimensional space, and then computing exact attention only for the top- k tokens. The quality of such a method rests entirely on one offline choice: the basis used to shrink queries and keys before scoring. Prior methods pick this basis by principal component analysis (PCA) of the keys, which minimizes key *reconstruction* error. We show that this is the wrong objective. The quantity that matters for top- k selection is the error in the *score* $\mathbf{q}^\top \mathbf{k}$, and we prove that, under a Gaussian surrogate, the expected squared score error decomposes into a cross-covariance *alignment* term plus a coupled *fluctuation* term. Key-PCA minimizes a degenerate case of the second term and ignores the first. We propose CrossCov-U, which builds a single shared orthonormal basis from the left singular vectors of the query-key cross-covariance $\mathbb{E}[\mathbf{k}\mathbf{q}^\top]$ and is the exact minimizer of the natural surrogate of the alignment term. For grouped-query attention we show that pooling the cross-covariance across the query heads of a group is the exact minimizer of the group-summed objective, so CrossCov-U needs no more bases than Loki and has identical runtime cost. Across [N] models, CrossCov-U raises attention-mass recall by Δ , lowers attention-output error by $\times$, and closes the perplexity gap to full attention from x to y at the same compression ratio, with [downstream accuracy summary].

1 Introduction

Autoregressive decoding with a large language model (LLM) must read the entire key-value (KV) cache at every generation step. At long context this cache can be larger than the model weights, and the cost of moving it from DRAM into registers dominates the attention step [Singhania et al., 2024, Ribar et al., 2024]. One popular way to reduce this cost without retraining is *sparse attention by selection*: compute cheap approximate scores for every cached token, pick the top- k tokens, and run exact attention only on those. Because no token is ever deleted, selection avoids the permanent information loss of eviction methods [Zhang et al., 2023, Xiao et al., 2024].

Loki [Singhania et al., 2024] made selection practical by observing that attention keys have low effective rank. It fits PCA to keys from a calibration set, rotates queries and keys into that basis, and scores tokens using only the first r of d coordinates. This works because PCA keeps the directions in which keys have the most energy, so the truncated keys stay close to the original keys.

This paper starts from a simple point that is easy to miss: *the selector never needs the keys to be close; it needs the scores to be close*. The score is a bilinear form $\mathbf{q}^\top \mathbf{k}$. A direction in key space that holds a lot of key energy contributes nothing to the score if queries never point along it. Conversely, a direction with little key energy can decide the ranking if queries consistently probe it. Key-PCA cannot see this distinction because it never looks at the queries. Figure 1 gives a two-dimensional example in which key-PCA collapses two keys that every query separates cleanly.

We make this intuition precise. We write down the expected squared score error of a rank- r shared-basis approximation and show that, under a joint-Gaussian surrogate, it splits into two parts (Theorem 4.5): a *residual alignment* term that depends only on the query-key cross-covariance $\mathbf{C} = \mathbb{E}[\mathbf{k}\mathbf{q}^\top]$, and a *coupled fluctuation* term that depends on the two auto-covariances through their product. Key-PCA minimizes the fluctuation term with the queries assumed isotropic and the alignment term dropped entirely. We then propose CrossCov-U, whose basis is the top- r left singular subspace of $\mathbf{C}$. It is the exact minimizer of the natural Frobenius surrogate of the alignment term (Theorem 4.9), it recovers Loki’s basis exactly when queries are replaced by keys (Lemma 4.13), and it preserves inner products exactly at full rank (Lemma 4.2).

[Figure 1. Left: 2-D toy. Two keys $\mathbf{k}_A, \mathbf{k}_B$ differ along a low-energy direction e_2 ; key-PCA keeps e_1 and maps them to the same point, while the query cloud lies along e_2 and separates them. The CrossCov-U axis (left singular vector of $\mathbb{E}[\mathbf{k}\mathbf{q}^\top]$) keeps e_2 . Right: overview of the pipeline (offline calibration $\rightarrow$ shared basis $\mathbf{U} \rightarrow$ rank- r latent scoring $\rightarrow$ top- $k \rightarrow$ exact attention).]

Figure 1: [Toy example and method overview.]

For grouped-query attention (GQA) we show that pooling $\mathbf{C}$ over the query heads in a group is the exact minimizer of the group-summed surrogate (Theorem 4.14), so CrossCov-U stores exactly one basis per KV head, the same as Loki.

CrossCov-U changes only the offline calibration statistic. The online algorithm, the memory layout, the kernels, and therefore the runtime cost are identical to Loki. Every accuracy gain we report is attributable to a single design choice.

Our contributions are:

- A decomposition of the expected squared score error of shared-basis sparse attention into an alignment term and a fluctuation term, with a mapping of Loki, SALS, SWAN and SVDq onto degenerate cases (Section 4).
- CrossCov-U: a closed-form basis from the cross-covariance $\mathbb{E}[\mathbf{k}\mathbf{q}^\top]$ that is surrogate-optimal for the alignment term, with a GQA-pooled construction that is exactly optimal for the group-summed surrogate (Section 5).
- A per-head *alignment ratio* ρ_h that predicts where CrossCov-U should beat key-PCA, and an offline oracle basis that minimizes the full objective, used to measure how much of the available gain CrossCov-U captures (Sections 7.3, 4.7).
- An evaluation on [models] showing higher attention-mass recall, lower attention-output error, and a smaller perplexity gap than Loki at matched compression, with [downstream tasks] (Section 7).

2 Background and Related Work

Attention. Consider one attention head with head dimension d . When generating token S , the head computes

$$\mathbf{y} = \text{softmax}\left(\frac{\mathbf{q}^\top \mathbf{K}^\top}{\sqrt{d}}\right) \mathbf{V}, \quad (1)$$

where $\mathbf{q} \in \mathbb{R}^d$ is the current query and $\mathbf{K}, \mathbf{V} \in \mathbb{R}^{S \times d}$ hold the keys and values of all S tokens seen so far. Modern models apply rotary position embeddings (RoPE) [Su et al., 2024] to $\mathbf{q}$ and $\mathbf{k}$ before the dot product.

We write $\mathbf{q}, \mathbf{k}$ for the vectors that actually enter the dot product and say explicitly when we mean pre-RoPE states.

Grouped-query attention. In GQA [Ainslie et al., 2023], a group of G query heads shares one key head and one value head. We index query heads in a group by $g \in \{1, \dots, G\}$ and write $\mathbf{q}^{(g)}$ for their queries; all of them score against the same $\mathbf{k}$.

Top- k selection. Selection methods compute an approximate score $\tilde{s}_j$ for every cached token j , keep the set $\mathcal{T}$ of the k largest, and compute (1) restricted to $\mathcal{T}$. Loki obtains $\tilde{s}_j$ by rotating $\mathbf{q}$ and $\mathbf{k}_j$ into a PCA basis of the keys and using only the first r coordinates, so the approximate scoring costs $O(Sr)$ instead of $O(Sd)$, and the exact part costs $O(kd)$.

2.1 Related Work

Low-rank keys and latent scoring. Loki [Singhanian et al., 2024] showed that attention keys have low effective rank across many models and datasets, and used a per-head key-PCA basis for latent scoring. SALS [SALS, 2025] stores keys in a joint-head key-PCA basis and reconstructs only the selected keys, adding memory compression. SWAN [SWAN, 2025] builds the basis from $\mathbb{E}[\mathbf{q}\mathbf{q}^\top] + \mathbb{E}[\mathbf{k}\mathbf{k}^\top]$. SVDq [SVDq, 2025] uses the key-PCA spectrum to allocate mixed precision across latent coordinates. All of these derive the basis from second moments of queries or keys separately; none uses the joint statistic $\mathbb{E}[\mathbf{k}\mathbf{q}^\top]$. Section 4 shows that each of them minimizes a degenerate case of the score-error objective.

Query-aware dimension selection. SparQ [Ribar et al., 2024] picks, per query, the coordinates where $|\mathbf{q}|$ is largest and scores keys on those coordinates. It is query-aware but works in the standard coordinate system and needs non-contiguous column gathers; it also keeps two copies of the keys. CrossCov-U is query-aware through calibration statistics, uses a fixed rotated basis, and keeps Loki’s contiguous slicing.

Cross-covariance in attention. Lambda Networks [Bello, 2021] contract positions first and express attention as a $d \times d$ matrix built from keys and values; the object $\mathbf{K}^\top \mathbf{Q}$ appears there as a *trained replacement* for attention. CrossCov-U uses $\mathbf{K}^\top \mathbf{Q}$ only as an offline calibration statistic for a post-hoc selector; the model is unchanged.

Eviction and compression. StreamingLLM [Xiao et al., 2024], H2O [Zhang et al., 2023], Scissorhands [Liu et al., 2023], KeyDiff [KeyDiff, 2025] and Ada-KV [Feng et al., 2025] permanently drop tokens. ShadowKV [Sun et al., 2024] keeps a low-rank pre-RoPE key store on GPU and exact values on CPU. Quantization of the KV cache [Liu et al., 2024] is complementary. CrossCov-U is a pure selection method: the full cache is retained and one dimension reduction is used for scoring only.

3 Why Key Energy Is the Wrong Target

Loki’s motivating analysis measures how much key variance the top- r principal components capture. We repeat that analysis and add the quantity that actually controls selection: how much *score* the same subspace captures. This section is empirical; Section 4 gives the theory.

3.1 Setup

Fix a layer and head. From a calibration set of N tokens we collect queries $\mathbf{Q} \in \mathbb{R}^{N \times d}$ and keys $\mathbf{K} \in \mathbb{R}^{N \times d}$ (both taken at the point where they enter the dot product; see Section 5.4 for pre- vs post-RoPE). We form three second-moment matrices:

$$\Sigma_k = \frac{1}{N} \mathbf{K}^\top \mathbf{K}, \quad \Sigma_q = \frac{1}{N} \mathbf{Q}^\top \mathbf{Q}, \quad \mathbf{C} = \frac{1}{N} \mathbf{K}^\top \mathbf{Q} \approx \mathbb{E}[\mathbf{k}\mathbf{q}^\top]. \quad (2)$$

Σ_k and Σ_q are symmetric positive semidefinite; $\mathbf{C}$ is square but generally not symmetric. Loki’s basis is the eigenbasis of Σ_k .

[Figure 2. For each layer (x-axis), the rank at which 90% of key energy is captured (Loki’s metric) and the rank at which 90% of score energy is captured, for the key-PCA basis and the CrossCov-U basis. Panels: Llama-2-7B, Mistral-7B, Llama-3.1-8B-Instruct. Pre- and post-RoPE as separate line styles.]

Figure 2: [Key-energy rank vs score-energy rank across layers.]

For any rank- r orthonormal $\mathbf{U} \in \mathbb{R}^{d \times r}$ we define two capture ratios:

$$\text{KeyEnergy}_r(\mathbf{U}) = \frac{\text{tr}(\mathbf{U}^\top \boldsymbol{\Sigma}_k \mathbf{U})}{\text{tr}(\boldsymbol{\Sigma}_k)}, \quad \text{ScoreEnergy}_r(\mathbf{U}) = \frac{\|\mathbf{U}^\top \mathbf{C}\|_F^2}{\|\mathbf{C}\|_F^2}. \quad (3)$$

The first is Loki’s “explained variance”. The second is the fraction of squared cross-covariance retained by $\mathbf{U}$; Theorem 4.9 shows it is the quantity a score-faithful basis should maximize.

3.2 Findings

We observe the following on [models] with [calibration sets].

1. **Key-PCA captures key energy but not score energy.** At the rank where key-PCA captures 90% of key energy, it captures only [x%] of score energy on average (Figure 2). The CrossCov-U basis captures [y%] at the same rank.
2. **The gap is head-dependent.** Heads with a large alignment ratio ρ_h (Figure 3) show the largest gap. These are the heads where queries and keys are systematically aligned, i.e., heads that do content-based retrieval.
3. **The structure is stable across calibration data.** The CrossCov-U basis fitted on [dataset A] captures [z%] of the score energy measured on [dataset B], matching the cross-dataset stability Loki reported for key-PCA.
4. **RoPE matters for both bases.** As in Loki, post-RoPE states have higher rank than pre-RoPE states. The fitted basis must be applied to states of the same kind it was fitted on (Section 5.4).

Definition 3.1 (Alignment ratio). For a head h with moments $\mathbf{C}_h, \boldsymbol{\Sigma}_q[, h], \boldsymbol{\Sigma}_k[, h]$, define

$$\rho_h = \frac{\|\mathbf{C}_h\|_F^2}{\text{tr}(\boldsymbol{\Sigma}_q[, h]) \text{tr}(\boldsymbol{\Sigma}_k[, h])/d}. \quad (4)$$

ρ_h compares the systematic query–key alignment with the product of the two total energies, normalized so that independent isotropic $\mathbf{q}, \mathbf{k}$ give $\rho_h \approx 1/d$ up to sampling noise. Theorem 4.5 explains why ρ_h predicts the advantage of CrossCov-U over key-PCA.

[Figure 3. Per-head alignment ratio ρ_h (Definition 3.1), shown as a heat map over (layer, KV head) for each model.]

Figure 3: [Alignment ratio ρ_h across layers and heads.]

4 Theory: Score-Faithful Bases

4.1 Setup and the exact objective

Let $\mathbf{U} \in \mathbb{R}^{d \times r}$ have orthonormal columns, $\mathbf{U}^\top \mathbf{U} = \mathbf{I}_r$. Define the projector onto its span and its complement,

$$\mathbf{P} = \mathbf{U}\mathbf{U}^\top, \quad \mathbf{M} = \mathbf{I} - \mathbf{P}, \quad \mathbf{P}^2 = \mathbf{P}, \mathbf{M}^2 = \mathbf{M}, \text{tr } \mathbf{M} = d - r. \quad (5)$$

A *shared-basis* approximation scores a token by

$$\tilde{s}(\mathbf{q}, \mathbf{k}) = (\mathbf{U}^\top \mathbf{q})^\top (\mathbf{U}^\top \mathbf{k}) = \mathbf{q}^\top \mathbf{P} \mathbf{k}, \quad (6)$$

so the score error is $s - \tilde{s} = \mathbf{q}^\top \mathbf{k} - \mathbf{q}^\top \mathbf{P} \mathbf{k} = \mathbf{q}^\top \mathbf{M} \mathbf{k}$. Loki, SALS, SWAN and CrossCov-U all have this form at inference time; they differ only in how $\mathbf{U}$ is chosen offline. The natural target is the expected squared score error over the joint distribution of $(\mathbf{q}, \mathbf{k})$ induced by the calibration corpus:

$$J(\mathbf{U}) = \mathbb{E}[(\mathbf{q}^\top \mathbf{M} \mathbf{k})^2]. \quad (7)$$

Remark 4.1 (Why J and not ranking loss). Top- k selection depends on the order of scores, not their values. But the ordering is a discontinuous function of the scores, and any bound on ranking error must go through the score error. J is the smallest smooth objective that controls it: a token with true score s_j and approximate score $\tilde{s}_j$ leaves the top- k set only if $|s_j - \tilde{s}_j|$ exceeds its margin to the k -th score. Section 7 shows empirically that lower J tracks higher attention-mass recall, lower attention-output error and lower perplexity.

4.2 Two elementary properties

Lemma 4.2 (Exactness at full rank; failure of the asymmetric variant). *Let $\mathbf{U} \in \mathbb{R}^{d \times d}$ be orthogonal. Then $\mathbf{q}^\top \mathbf{P} \mathbf{k} = \mathbf{q}^\top \mathbf{k}$ for all $\mathbf{q}, \mathbf{k}$. In contrast, for any two orthogonal $\mathbf{U}, \mathbf{V} \in \mathbb{R}^{d \times d}$, the asymmetric score $(\mathbf{V}^\top \mathbf{q})^\top (\mathbf{U}^\top \mathbf{k}) = \mathbf{q}^\top \mathbf{V} \mathbf{U}^\top \mathbf{k}$ equals $\mathbf{q}^\top \mathbf{k}$ for all $\mathbf{q}, \mathbf{k}$ if and only if $\mathbf{V} = \mathbf{U}$.*

Proof. If $\mathbf{U}$ is square and orthogonal then $\mathbf{P} = \mathbf{U}\mathbf{U}^\top = \mathbf{I}$ and the first claim is immediate. For the second, $\mathbf{q}^\top \mathbf{V} \mathbf{U}^\top \mathbf{k} = \mathbf{q}^\top \mathbf{k}$ for all $\mathbf{q}, \mathbf{k}$ means $\mathbf{V} \mathbf{U}^\top = \mathbf{I}$, i.e. $\mathbf{V} = \mathbf{U}$ since $\mathbf{U}$ is orthogonal. Otherwise $\mathbf{V} \mathbf{U}^\top$ is a non-identity rotation and there exist $\mathbf{q}, \mathbf{k}$ with $\mathbf{q}^\top \mathbf{V} \mathbf{U}^\top \mathbf{k} \neq \mathbf{q}^\top \mathbf{k}$. $\square$

Lemma 4.2 explains a negative result we obtained early on. The singular value decomposition $\mathbf{C} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$ offers two natural factors, and the tempting design is to project keys with $\mathbf{U}$ and queries with $\mathbf{V}$. This does not converge to exact attention as $r \rightarrow d$, because the full-rank map $\mathbf{V} \mathbf{U}^\top$ is a rotation, not the identity.

Experimentally its recall sat at the random-guessing floor. The deployable object is the *shared* left singular subspace, for which $\mathbf{P} \rightarrow \mathbf{I}$ as $r \rightarrow d$.

Lemma 4.3 (Pointwise score-error bound). *For any $\mathbf{q}, \mathbf{k}$ and projector $\mathbf{M}$, $|\mathbf{q}^\top \mathbf{M} \mathbf{k}| \leq \|\mathbf{M} \mathbf{q}\| \|\mathbf{M} \mathbf{k}\|$.*

Proof. $\mathbf{q}^\top \mathbf{M} \mathbf{k} = (\mathbf{M} \mathbf{q})^\top (\mathbf{M} \mathbf{k})$ since $\mathbf{M} = \mathbf{M}^\top = \mathbf{M}^2$; apply Cauchy–Schwarz. $\square$

Lemma 4.3 is the analogue of Loki’s Lemma 4.2. Key-PCA minimizes $\mathbb{E} \|\mathbf{M} \mathbf{k}\|^2 = \text{tr}(\mathbf{M} \boldsymbol{\Sigma}_k)$, i.e. the second factor only. The bound is loose in exactly the situation of Figure 1: the residual key is large but orthogonal to every residual query.

4.3 Decomposition of the objective

Assumption 4.4 (Gaussian surrogate). $(\mathbf{q}, \mathbf{k})$ are jointly Gaussian with zero mean, cross-covariance $\mathbf{C} = \mathbb{E}[\mathbf{k} \mathbf{q}^\top]$ and auto-covariances $\boldsymbol{\Sigma}_q = \mathbb{E}[\mathbf{q} \mathbf{q}^\top]$, $\boldsymbol{\Sigma}_k = \mathbb{E}[\mathbf{k} \mathbf{k}^\top]$.

The zero-mean part is a convenience; Remark 4.6 handles means. The Gaussian part is used only to close the fourth moment; the first term of the decomposition below holds without it.

Theorem 4.5 (Alignment–fluctuation decomposition). *Under Assumption 4.4,*

$$J(\mathbf{U}) = \underbrace{(\text{tr} \mathbf{M} \mathbf{C})^2 + \text{tr}(\mathbf{M} \mathbf{C} \mathbf{M} \mathbf{C})}_{J_{\text{align}}(\mathbf{U})} + \underbrace{\text{tr}(\mathbf{M} \boldsymbol{\Sigma}_q \mathbf{M} \boldsymbol{\Sigma}_k)}_{J_{\text{fluc}}(\mathbf{U})}. \quad (8)$$

Without the Gaussian assumption, $J(\mathbf{U}) = (\text{tr} \mathbf{M} \mathbf{C})^2 + \text{Var}(\mathbf{q}^\top \mathbf{M} \mathbf{k})$ still holds; only the expression for the variance uses Gaussianity.

Proof. Write $\mathbf{q}^\top \mathbf{M} \mathbf{k} = \sum_{i,j} M_{ij} q_i k_j$. Then

$$J = \sum_{i,j,a,b} M_{ij} M_{ab} \mathbb{E}[q_i k_j q_a k_b].$$

By Isserlis’ theorem for zero-mean Gaussians, $\mathbb{E}[q_i k_j q_a k_b] = \mathbb{E}[q_i k_j] \mathbb{E}[q_a k_b] + \mathbb{E}[q_i q_a] \mathbb{E}[k_j k_b] + \mathbb{E}[q_i k_b] \mathbb{E}[q_a k_j]$. Using $C_{ji} = \mathbb{E}[k_j q_i]$, $(\boldsymbol{\Sigma}_q)_{ia} = \mathbb{E}[q_i q_a]$, $(\boldsymbol{\Sigma}_k)_{jb} = \mathbb{E}[k_j k_b]$, the three pairings give:

$$\begin{aligned} \sum_{i,j} M_{ij} C_{ji} \sum_{a,b} M_{ab} C_{ba} &= (\text{tr} \mathbf{M} \mathbf{C})^2, \\ \sum_{i,j,a,b} M_{ij} (\boldsymbol{\Sigma}_q)_{ia} M_{ab} (\boldsymbol{\Sigma}_k)_{jb} &= \sum_{j,b} (\mathbf{M}^\top \boldsymbol{\Sigma}_q \mathbf{M})_{jb} (\boldsymbol{\Sigma}_k)_{jb} = \text{tr}(\mathbf{M} \boldsymbol{\Sigma}_q \mathbf{M} \boldsymbol{\Sigma}_k), \\ \sum_{i,j,a,b} M_{ij} C_{bi} M_{ab} C_{ja} &= \sum_{i,a} \left(\sum_j M_{ij} C_{ja} \right) \left(\sum_b M_{ab} C_{bi} \right) = \sum_{i,a} (\mathbf{M} \mathbf{C})_{ia} (\mathbf{M} \mathbf{C})_{ai} = \text{tr}(\mathbf{M} \mathbf{C} \mathbf{M} \mathbf{C}), \end{aligned}$$

where we used $\mathbf{M} = \mathbf{M}^\top$ and $\boldsymbol{\Sigma}_k = \boldsymbol{\Sigma}_k^\top$ in the second line. For the non-Gaussian statement, $\mathbb{E}[\mathbf{q}^\top \mathbf{M} \mathbf{k}] = \text{tr}(\mathbf{M} \mathbf{C})$ and $\mathbb{E}[X^2] = (\mathbb{E}X)^2 + \text{Var}X$. $\square$

Remark 4.6 (Non-zero means). If $\mathbf{q} = \bar{\mathbf{q}} + \mathbf{q}'$, $\mathbf{k} = \bar{\mathbf{k}} + \mathbf{k}'$ with centered $\mathbf{q}', \mathbf{k}'$, then $\mathbf{q}^\top \mathbf{M} \mathbf{k} = \bar{\mathbf{q}}^\top \mathbf{M} \bar{\mathbf{k}} + \bar{\mathbf{q}}^\top \mathbf{M} \mathbf{k}' + \mathbf{q}'^\top \mathbf{M} \bar{\mathbf{k}} + \mathbf{q}'^\top \mathbf{M} \mathbf{k}'$. The uncentered second moment $\mathbb{E}[\mathbf{k} \mathbf{q}^\top] = \bar{\mathbf{k}} \bar{\mathbf{q}}^\top + \mathbf{C}'$ that we compute in practice therefore already includes the mean product, and the extra cross terms are of the form $\bar{\mathbf{q}}^\top \mathbf{M} \mathbf{k}'$, which the Frobenius surrogate of Section 4.5 also controls since $\mathbb{E}(\bar{\mathbf{q}}^\top \mathbf{M} \mathbf{k}')^2 = \bar{\mathbf{q}}^\top \mathbf{M} \boldsymbol{\Sigma}_k' \mathbf{M} \bar{\mathbf{q}} \leq \|\bar{\mathbf{q}}\|^2 \text{tr}(\mathbf{M} \boldsymbol{\Sigma}_k' \mathbf{M})$. We use uncentered moments throughout.

Interpretation. J_{align} is the systematic query–key correlation left in the discarded subspace: if keys and queries are consistently aligned along a direction, throwing that direction away biases every score. J_{fluc} is residual query energy times residual key energy, coupled through the shared discarded directions; it is the error one would see even with no systematic alignment.

4.4 Prior methods as degenerate cases

Proposition 4.7 (What prior bases minimize). *Over rank- r projectors $\mathbf{P}$:*

- (a) *Key-PCA (Loki, SALS, Palu, SVDq) minimizes $\text{tr}(\mathbf{M}\mathbf{\Sigma}_k)$, which equals J_{fluc} when $\mathbf{\Sigma}_q = \mathbf{I}$; it ignores J_{align} entirely. Equivalently it minimizes J under the two false assumptions $\mathbf{C} = \mathbf{0}$ and isotropic queries.*
- (b) *SWAN minimizes $\text{tr}(\mathbf{M}\mathbf{\Sigma}_q) + \text{tr}(\mathbf{M}\mathbf{\Sigma}_k)$, an additive surrogate of the multiplicative J_{fluc} ; it also ignores J_{align} .*
- (c) *CrossCov-U minimizes $\|\mathbf{MC}\|_F^2$, a surrogate that upper bounds J_{align} (Proposition 4.10).*

Proof. (a) With $\mathbf{\Sigma}_q = \mathbf{I}$, $\text{tr}(\mathbf{M}\mathbf{I}\mathbf{M}\mathbf{\Sigma}_k) = \text{tr}(\mathbf{M}^2\mathbf{\Sigma}_k) = \text{tr}(\mathbf{M}\mathbf{\Sigma}_k)$ because $\mathbf{M}^2 = \mathbf{M}$. Loki’s basis is the top- r eigenbasis of $\mathbf{\Sigma}_k$, which is the minimizer of $\text{tr}(\mathbf{M}\mathbf{\Sigma}_k)$ by the Ky Fan inequality. (b) and (c) are the stated objectives of the respective methods; see Section 4.5 for (c). $\square$

Proposition 4.8 (Product versus sum). *Suppose $\mathbf{\Sigma}_q$ and $\mathbf{\Sigma}_k$ commute, with common eigenbasis and eigenvalues $\{a_i\}$, $\{b_i\}$. Among projectors diagonal in that basis that discard a coordinate set D with $|D| = d - r$,*

$$J_{\text{fluc}} = \text{tr}(\mathbf{M}\mathbf{\Sigma}_q\mathbf{M}\mathbf{\Sigma}_k) = \sum_{i \in D} a_i b_i. \quad (9)$$

The exact minimizer discards the coordinates with the smallest products $a_i b_i$; SWAN discards the smallest sums $a_i + b_i$; key-PCA discards the smallest b_i . These disagree exactly when the spectra are imbalanced: a direction with large query energy but negligible key energy is kept by SWAN and correctly discarded by the product rule.

Proof. In the common eigenbasis $\mathbf{M} = \text{diag}(m_i)$ with $m_i \in \{0, 1\}$, $\mathbf{\Sigma}_q = \text{diag}(a_i)$, $\mathbf{\Sigma}_k = \text{diag}(b_i)$, so $\text{tr}(\mathbf{M}\mathbf{\Sigma}_q\mathbf{M}\mathbf{\Sigma}_k) = \sum_i m_i^2 a_i b_i = \sum_{i \in D} a_i b_i$. $\square$

The restriction to axis-aligned projectors is for illustration; the unconstrained minimizer of J_{fluc} need not be axis-aligned.

4.5 The CrossCov-U basis and its optimality

Theorem 4.9 (Eckart–Young for the alignment surrogate). *Over rank- r orthonormal $\mathbf{U}$, the surrogate*

$$\hat{J}(\mathbf{U}) = \|\mathbf{MC}\|_F^2 = \|(\mathbf{I} - \mathbf{U}\mathbf{U}^\top)\mathbf{C}\|_F^2 \quad (10)$$

is minimized by $\mathbf{U}^$ = the top- r left singular vectors of $\mathbf{C}$, equivalently the top- r eigenvectors of $\mathbf{C}\mathbf{C}^\top$. The minimum value is $\sum_{i > r} \sigma_i(\mathbf{C})^2$.*

Proof. $\|\mathbf{MC}\|_F^2 = \text{tr}(\mathbf{C}^\top \mathbf{M}^2 \mathbf{C}) = \text{tr}(\mathbf{C}^\top \mathbf{C}) - \text{tr}(\mathbf{C}^\top \mathbf{P} \mathbf{C}) = \|\mathbf{C}\|_F^2 - \text{tr}(\mathbf{U}^\top \mathbf{C} \mathbf{C}^\top \mathbf{U})$. Minimizing $\hat{J}$ is therefore maximizing $\text{tr}(\mathbf{U}^\top \mathbf{C} \mathbf{C}^\top \mathbf{U})$ over orthonormal $\mathbf{U}$. By the Ky Fan maximum principle this is at most the sum of the r largest eigenvalues of $\mathbf{C} \mathbf{C}^\top$, attained when the columns of $\mathbf{U}$ span the corresponding eigenspace. The eigenvectors of $\mathbf{C} \mathbf{C}^\top$ are the left singular vectors of $\mathbf{C}$ and its eigenvalues are $\sigma_i(\mathbf{C})^2$, which gives the minimum value. $\square$

Proposition 4.10 (The surrogate controls both alignment terms). *For any projector $\mathbf{M}$ of rank $d - r$,*

$$|\text{tr}(\mathbf{M}\mathbf{C}\mathbf{M}\mathbf{C})| \leq \|\mathbf{MC}\|_F^2, \quad (\text{tr} \mathbf{MC})^2 \leq (d - r) \|\mathbf{MC}\|_F^2, \quad (11)$$

hence $J_{\text{align}} \leq (d - r + 1) \|\mathbf{MC}\|_F^2$, and $\mathbf{U}^$ of Theorem 4.9 minimizes this bound exactly.*

Proof. First inequality: $\text{tr}(\mathbf{M}\mathbf{C}\mathbf{M}\mathbf{C}) = \text{tr}((\mathbf{MC})(\mathbf{MC})) = \langle (\mathbf{MC})^\top, \mathbf{MC} \rangle_F \leq \|(\mathbf{MC})^\top\|_F \|\mathbf{MC}\|_F = \|\mathbf{MC}\|_F^2$ by the trace Cauchy–Schwarz inequality. Second: $\text{tr}(\mathbf{MC}) = \text{tr}(\mathbf{M} \cdot \mathbf{MC}) = \langle \mathbf{M}, \mathbf{MC} \rangle_F$, and $\|\mathbf{M}\|_F^2 = \text{tr} \mathbf{M} = d - r$, so $(\text{tr} \mathbf{MC})^2 \leq \|\mathbf{M}\|_F^2 \|\mathbf{MC}\|_F^2 = (d - r) \|\mathbf{MC}\|_F^2$. $\square$

Proposition 4.11 (Exact optimality under isotropic fluctuations). *If $\Sigma_q = \sigma_q^2 \mathbf{I}$ and $\Sigma_k = \sigma_k^2 \mathbf{I}$ then $J_{\text{fluc}} = \sigma_q^2 \sigma_k^2 (d - r)$ is constant in $\mathbf{U}$, so minimizing J is minimizing J_{align} alone. If in addition $\mathbf{C}$ is symmetric positive semidefinite, the top- r eigenbasis of $\mathbf{C}$ minimizes J exactly.*

Proof. $\text{tr}(\mathbf{M} \sigma_q^2 \mathbf{I} \mathbf{M} \sigma_k^2 \mathbf{I}) = \sigma_q^2 \sigma_k^2 \text{tr}(\mathbf{M}^2) = \sigma_q^2 \sigma_k^2 (d - r)$. If $\mathbf{C} = \mathbf{C}^\top \succeq 0$ with eigenvalues $c_1 \geq \dots \geq c_d \geq 0$, let $\mu_1 \geq \dots \geq \mu_{d-r}$ be the eigenvalues of the compression $\mathbf{MCM}$ restricted to $\text{range}(\mathbf{M})$. Then $\text{tr}(\mathbf{MC}) = \sum_i \mu_i$ and, using $\mathbf{M}^2 = \mathbf{M}$ and cyclicity, $\text{tr}(\mathbf{MCMC}) = \text{tr}((\mathbf{MCM})^2) = \sum_i \mu_i^2$. By Cauchy interlacing, $\mu_i \geq c_{r+i}$ for every i , so both sums are at least their values at the axis-aligned choice that discards the $d - r$ smallest eigen-directions, and that choice attains them. Hence the top- r eigenbasis of $\mathbf{C}$ minimizes J_{align} , and since $\mathbf{C}$ is symmetric PSD its left singular vectors are its eigenvectors, so this is the CrossCov-U basis. $\square$

Corollary 4.12 (Regime prediction). *When $\|\mathbf{C}\|_F^2$ is large relative to $\text{tr}(\Sigma_q) \text{tr}(\Sigma_k)/d$ (large ρ_h), J_{align} dominates J and the CrossCov-U basis is near-optimal. When ρ_h is small, J_{fluc} dominates and the optimal basis moves toward coupled-auto-covariance structure. The advantage of CrossCov-U over key-PCA is therefore predicted to increase with ρ_h .*

Lemma 4.13 (Loki is the self-collapse of CrossCov-U). *If queries are replaced by keys in the calibration statistic, i.e. $\mathbf{C} \leftarrow \mathbb{E}[\mathbf{k}\mathbf{k}^\top] = \Sigma_k$, then the CrossCov-U basis equals the key-PCA basis.*

Proof. Σ_k is symmetric PSD, so its left singular vectors are its eigenvectors ordered by eigenvalue, which is the PCA basis. $\square$

Lemma 4.13 is a useful unit test: pointing the calibration code at key tensors on both sides must reproduce Loki’s components bit-for-bit.

4.6 Grouped-query attention

Under GQA, one key head serves G query heads, so there are G cross-covariances $\mathbf{C}_g = \mathbb{E}[\mathbf{k}\mathbf{q}^{(g)\top}]$ per KV head. Loki stores one basis per KV head. A fair comparison must do the same.

Theorem 4.14 (GQA pooling is exactly surrogate-optimal). *Define the group-summed surrogate $\hat{J}_{\text{grp}}(\mathbf{U}) = \sum_{g=1}^G \|\mathbf{MC}_g\|_F^2$. It is minimized by the top- r eigenvectors of $\mathbf{A} = \sum_{g=1}^G \mathbf{C}_g \mathbf{C}_g^\top$, equivalently the top- r left singular vectors of the horizontally stacked matrix $[\mathbf{C}_1 | \mathbf{C}_2 | \dots | \mathbf{C}_G] \in \mathbb{R}^{d \times Gd}$.*

Proof. $\sum_g \|\mathbf{MC}_g\|_F^2 = \|\mathbf{M}[\mathbf{C}_1 | \dots | \mathbf{C}_G]\|_F^2$ since the Frobenius norm of a block-row matrix is the sum over blocks. Apply Theorem 4.9 to the stacked matrix; its left singular vectors are the eigenvectors of $[\mathbf{C}_1 | \dots | \mathbf{C}_G][\mathbf{C}_1 | \dots | \mathbf{C}_G]^\top = \sum_g \mathbf{C}_g \mathbf{C}_g^\top$. $\square$

This promotes GQA pooling from a heuristic to the exact solution of the group-summed problem, and it gives CrossCov-U the same basis count and storage as Loki. It also contrasts with SALS’s joint-head lemma, which argues only that stacking key auto-covariances across heads captures at least as much key energy; Theorem 4.14 is a statement about score error and respects the GQA score structure (each $\mathbf{q}^{(g)}$ really does score against the same $\mathbf{k}$).

Remark 4.15 (Per-query-head upper bound). Fitting a separate basis $\mathbf{U}_g$ from each $\mathbf{C}_g$ minimizes each $\|\mathbf{M}_g \mathbf{C}_g\|_F^2$ individually and therefore lower-bounds $\hat{J}_{\text{grp}}$; but it requires G latent copies of every key. We report it as an oracle to show how much the shared constraint costs (Section 7).

4.7 Pre-RoPE closed form and the oracle basis

Before RoPE, $\mathbf{q}^{(g)} = \mathbf{W}_q^{(g)} \mathbf{x}$ and $\mathbf{k} = \mathbf{W}_k \mathbf{x}$ are linear in the same hidden state $\mathbf{x} \in \mathbb{R}^{d_{\text{model}}}$ with second moment $\Sigma_x = \mathbb{E}[\mathbf{x}\mathbf{x}^\top]$. Then

$$\begin{aligned} \mathbf{C}_g &= \mathbf{W}_k \Sigma_x \mathbf{W}_q^{(g)\top}, & \Sigma_q^{(g)} &= \mathbf{W}_q^{(g)} \Sigma_x \mathbf{W}_q^{(g)\top}, \\ \Sigma_k &= \mathbf{W}_k \Sigma_x \mathbf{W}_k^\top, & \mathbf{A} &= \mathbf{W}_k \Sigma_x \left(\sum_g \mathbf{W}_q^{(g)\top} \mathbf{W}_q^{(g)} \right) \Sigma_x \mathbf{W}_k^\top. \end{aligned} \quad (12)$$

Every term of (8) is therefore a known function of the weights and a single $d_{\text{model}} \times d_{\text{model}}$ calibration statistic. This has two uses.

Algorithm 1 CrossCov-U offline calibration (one layer, one KV head)

Require: calibration states $\mathbf{K} \in \mathbb{R}^{N \times d}$, $\mathbf{Q}^{(1..G)} \in \mathbb{R}^{N \times d}$ taken at the point where they enter the dot product

```
1:  $\mathbf{A} \leftarrow \mathbf{0}_{d \times d}$ 
2: for  $g = 1, \dots, G$  do
3:    $\mathbf{C}_g \leftarrow \frac{1}{N} \mathbf{K}^\top \mathbf{Q}^{(g)}$   $\triangleright$  cross-covariance,  $d \times d$ 
4:    $\mathbf{A} \leftarrow \mathbf{A} + \mathbf{C}_g \mathbf{C}_g^\top$ 
5: end for
6:  $(\mathbf{\Lambda}, \mathbf{U}_{\text{full}}) \leftarrow \text{eigh}(\mathbf{A})$ , columns sorted by descending eigenvalue
7: assert  $\mathbf{U}_{\text{full}}^\top \mathbf{U}_{\text{full}} = \mathbf{I}$  (orthonormality)
8: emit  $\mathbf{U}_{\text{full}}$  and metadata {rotary type, calibration set,  $G$ }
```

Algorithm 2 CrossCov-U online step (one query head; identical to Loki’s)

Require: basis $\mathbf{U}_{\text{full}}$ of the head’s KV group, scoring rank r , budget k

Require: rotated key cache $\hat{\mathbf{K}} = \mathbf{K} \mathbf{U}_{\text{full}} \in \mathbb{R}^{S \times d}$ (maintained incrementally), values $\mathbf{V}$

```
1:  $\hat{\mathbf{q}} \leftarrow \mathbf{U}_{\text{full}}^\top \mathbf{q}$   $\triangleright O(d^2)$ 
2:  $\tilde{\mathbf{s}} \leftarrow \hat{\mathbf{K}}_{:,1:r} \hat{\mathbf{q}}_{1:r}$   $\triangleright$  latent scores,  $O(Sr)$ 
3:  $\mathcal{T} \leftarrow \text{topk}(\tilde{\mathbf{s}}, k)$ 
4:  $\mathbf{s}_{\mathcal{T}} \leftarrow \hat{\mathbf{K}}_{\mathcal{T},:} \hat{\mathbf{q}} / \sqrt{d}$   $\triangleright$  exact scores,  $O(kd)$ , by Lemma 4.2
5:  $\mathbf{y} \leftarrow \text{softmax}(\mathbf{s}_{\mathcal{T}}) \mathbf{V}_{\mathcal{T},:}$   $\triangleright O(kd)$ 
```

Oracle basis. The full group-summed objective $J_{\text{grp}}(\mathbf{U}) = \sum_g [\text{tr}(\mathbf{M} \mathbf{C}_g)^2 + \text{tr}(\mathbf{M} \mathbf{C}_g \mathbf{M} \mathbf{C}_g) + \text{tr}(\mathbf{M} \mathbf{\Sigma}_q^{(g)} \mathbf{M} \mathbf{\Sigma}_k)]$ is a smooth function on the Grassmannian $\text{Gr}(r, d)$. At $d = 128$ it can be minimized offline per KV head by Riemannian gradient descent (Appendix B) in seconds. We call the result the *J-opt oracle*. It is deployable (it is just another fixed basis), and it lets us report what fraction of the Loki→oracle gap each closed-form basis closes. If CrossCov-U closes most of it, the closed form is the right headline; if not, the oracle itself is the method.

Sanity check of the surrogate. Because (12) needs only $\mathbf{\Sigma}_x$, we can evaluate J for any candidate basis and check that the Gaussian-surrogate ranking of bases agrees with the empirical score $\text{MSE} \frac{1}{N} \sum_n (\mathbf{q}_n^\top \mathbf{M} \mathbf{k}_n)^2$ and with downstream recall. Section 7 reports this agreement.

5 CrossCov-U: Algorithm and Cost

CrossCov-U has the same two phases as Loki. Only the statistic diagonalized in the offline phase changes.

5.1 Offline calibration

For each layer and each KV head we run the model on a calibration corpus of N tokens (Section 6) and accumulate, for each query head g in the group, $\mathbf{C}_g = \frac{1}{N} \mathbf{K}^\top \mathbf{Q}^{(g)}$. We form $\mathbf{A} = \sum_g \mathbf{C}_g \mathbf{C}_g^\top$, take its eigendecomposition $\mathbf{A} = \mathbf{U}_{\text{full}} \mathbf{\Lambda} \mathbf{U}_{\text{full}}^\top$ with eigenvalues in descending order, and store the complete orthogonal matrix $\mathbf{U}_{\text{full}} \in \mathbb{R}^{d \times d}$ (Algorithm 1). Storing all d columns, as Loki does, keeps the online rotation exact (Lemma 4.2): the first r columns give the latent scores and the full rotation gives exact scores on the selected tokens.

5.2 Online selection

Algorithm 2 is Loki’s algorithm verbatim. Each new key is rotated once when written to the cache; each query is rotated once per step. Latent scores use the first r rotated coordinates of every cached key; exact scores on the top- k use all d rotated coordinates. Because $\mathbf{U}_{\text{full}}$ is orthogonal, the exact scores equal $\mathbf{q}^\top \mathbf{k}_j$ and no reconstruction is needed. Values are untouched.

Table 1: Per-head, per-step cost. CrossCov-U and Loki are identical by construction.

	Elements read	Multiply-adds	Extra state
Full attention	$2Sd$	$2Sd$	–
Loki	$Sr + 2kd$	$Sr + 2kd + 2d^2$	$\mathbf{U}_{\text{full}}$ (d^2 per KV head)
CrossCov-U	$Sr + 2kd$	$Sr + 2kd + 2d^2$	$\mathbf{U}_{\text{full}}$ (d^2 per KV head)

5.3 Cost analysis

Let S be the cache length, d the head dimension, r the scoring rank and k the number of selected tokens, and count multiply-adds and bytes moved per query head per decode step (Table 1). Full attention reads $2Sd$ elements and does $2Sd$ multiply-adds. Loki and CrossCov-U read $Sr + 2kd$ elements and do $Sr + 2kd$ multiply-adds plus $2d^2$ for the two rotations, which is independent of S . With Loki’s standard settings $r/d = 1/4$ and $k/S = 1/8$ the data-movement ratio is $(r/d + 2k/S)/2 = (0.25 + 0.25)/2 = 1/4$.

Because the two methods share layouts and kernels, we do not claim any throughput advantage over Loki, and we reuse Loki’s Triton kernels unchanged. The systems claim of this paper is that the selection family delivers real decode speedup at long context (already established by Loki); the accuracy claim is that CrossCov-U achieves better recall and perplexity at the same compression ratio. We keep these two claims separate.

5.4 Handling rotary embeddings

RoPE is a position-dependent rotation applied to $\mathbf{q}$ and $\mathbf{k}$ before the dot product. Two design points follow.

Fit/inference alignment is mandatory. If the basis is fitted on pre-RoPE states, it must be applied to pre-RoPE states; if fitted post-RoPE, applied post-RoPE. A mismatch destroys recall (Section 7; we observed near-random recall in an early run with a pre-RoPE basis applied to post-RoPE keys). Algorithm 1 records the rotary type in the emitted metadata and the loader refuses a mismatch.

Post-RoPE is the default for selection. Post-RoPE $\mathbf{C}$ averages the rotation over positions in the calibration set. Its top singular directions are therefore the directions along which queries and keys stay aligned *after* RoPE, which is what the deployed dot product sees. The price is a higher effective rank, as Loki also found. The pre-RoPE alternative, used by SALS and ShadowKV for storage, keeps rank low but requires applying RoPE after selection and raises the question of whether pre-RoPE subspaces transfer across sequences at long context. We isolate the two factors with the 2×2 experiment $\{\text{pre}, \text{post}\} \times \{\Sigma_k, \mathbf{C}\}$ in Section 7.

5.5 What CrossCov-U is not

The CrossCov-U basis is a *selection* basis, not a *storage* basis. Storing only the rank- r latents $\hat{\mathbf{K}}_{:,1:r}$ and using the latent scores as final attention logits fails badly: since $\mathbf{q}^\top \mathbf{P} \mathbf{k} = (\mathbf{U}^\top \mathbf{q})^\top (\mathbf{U}^\top \mathbf{k})$, this is exactly the selection score used as the final logit, and $\mathbf{U}$ was chosen to capture alignment directions, not key energy, so peak logits shrink and the softmax flattens. Softmax is sensitive to score *magnitudes*, not just their order. If memory compression is wanted, the correct design is two bases: store in a key-energy basis (Loki/SALS) and select with CrossCov-U; we leave this to future work.

6 Experimental Setup

Models. [Llama-2-7B, Mistral-7B-v0.2 (GQA, $G = 4$), Llama-3.1-8B-Instruct (GQA, $G = 4$), OpenLLaMA-3B-v2; add larger models if compute allows.]

Table 2: Selection fidelity at $k/S = 1/8$ versus scoring rank r . Higher recall and lower relL2 are better. [Fill.]

Model	Method	Mass recall			relL2		
		$r=8$	$r=16$	$r=32$	$r=8$	$r=16$	$r=32$
Mistral-7B	Loki	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx
	SALS-basis	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx
	SWAN	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx
	CrossCov-U	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx
Llama-3.1-8B-Instruct	Loki	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx
	SALS-basis	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx
	SWAN	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx
	CrossCov-U	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx
Llama-2-7B	Loki	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx
	CrossCov-U	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx

Calibration. [WikiText-2 train split (and C4/BookCorpus for the cross-dataset test), N tokens, sequence length L , chunking; state whether moments are uncentered (they are).]

Baselines. Full attention; Loki (per-KV-head key-PCA) [Singhania et al., 2024]; SALS basis (joint-head key-PCA, used as a selector) [SALS, 2025]; SWAN basis [SWAN, 2025]; SparQ [Ribar et al., 2024]; H2O [Zhang et al., 2023] as an eviction reference; the per-query-head oracle of Section 4.6; and the J -opt oracle of Section 4.7. All selection baselines use the same r , k , sink/recent protection and kernels.

Metrics.

- *Attention-mass recall*: the fraction of full-attention probability mass that falls on the selected set $\mathcal{T}$, averaged over heads and positions.
- *Attention-output error* (relL2): $\|\mathbf{y}_{\text{sparse}} - \mathbf{y}_{\text{full}}\| / \|\mathbf{y}_{\text{full}}\|$.
- *Perplexity* on WikiText-2 [and C4], at sequence length $[L]$.
- *Downstream accuracy*: [lm-evaluation-harness tasks (as in Loki: HellaSwag, MMLU, Winogrande, ARC, TruthfulQA, GSM8K) and long-context tasks (LongBench, RULER, HotpotQA, Musique)].
- *Score MSE and surrogate*: empirical $\frac{1}{N} \sum_n (\mathbf{q}_n^\top \mathbf{M} \mathbf{k}_n)^2$ and the closed-form J of (8), per head.

Compression settings. Following Loki we sweep $r/d \in \{1/16, 1/8, 1/4, 1/2\}$ and $k/S \in \{1/8, 1/4, 1/2\}$. Unless stated otherwise, headline numbers use $r/d = 1/4$, $k/S = 1/8$. [State d for each model and the exact r values.]

Hardware. [H100 80GB; Nsight Compute for DRAM-byte measurements.]

7 Results

7.1 Selection fidelity

[One paragraph: CrossCov-U improves recall by Δ and cuts relL2 by $\times$ at every rank; the gap is largest at small r where the basis choice matters most and closes as $r \rightarrow d$ (Lemma 4.2).]

Table 3: Perplexity (WikiText-2) and downstream accuracy at $r/d = 1/4$, $k/S = 1/8$, in the format of Loki’s Table 1. [Fill; add rows for other (r, k) settings in the appendix.]

Model	Method	PPL	HellaSwag	MMLU	Winogrande	ARC-c	TruthfulQA	GSM8K	Avg
[Model 1]	Full	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx
	H2O	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx
	Loki	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx
	CrossCov-U	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx
[Model 2]	Full	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx
	H2O	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx
	Loki	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx
	CrossCov-U	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx	x.xx

Table 4: Long-context tasks at $r/d = 1/4$, $k/S = 1/8$. [Fill; RULER subtasks are where SALS-type selectors are known to crack, so report them separately.]

Model	Method	LongBench (avg)	RULER (avg)	HotpotQA	Musique
[Model]	Full	x.xx	x.xx	x.xx	x.xx
	Loki	x.xx	x.xx	x.xx	x.xx
	CrossCov-U	x.xx	x.xx	x.xx	x.xx

7.2 Perplexity and downstream accuracy

[One paragraph. Include the causal chain: better basis $\rightarrow$ higher recall $\rightarrow$ lower relL2 $\rightarrow$ smaller PPL gap $\rightarrow$ smaller accuracy drop. State the average accuracy degradation vs full attention for both methods, as Loki did (“only 6.8%”).]

7.3 Ablations

Table 5: Basis construction under GQA (Mistral-7B, $G = 4$, $k/S = 1/8$). Per-query-head is an oracle that needs G latent key copies. [Fill.]

Basis	#bases/layer	Recall $r=8$	Recall $r=16$	relL2 $r=8$
Loki (key-PCA, per KV head)	[8]	x.xx	x.xx	x.xx
CrossCov-U, first query head only	[8]	x.xx	x.xx	x.xx
CrossCov-U, pooled (Theorem 4.14)	[8]	x.xx	x.xx	x.xx
CrossCov-U, per query head (oracle)	[32]	x.xx	x.xx	x.xx

GQA pooling.

Alignment ratio predicts the gain. [Report Spearman ρ and note that low- ρ_h heads show no loss, consistent with Corollary 4.12.]

Oracle gap. [Also report the rank correlation between closed-form J and empirical score MSE across bases and heads, validating the Gaussian surrogate.]

Rotary treatment (2×2).

Cross-dataset generalization. [Table: basis fitted on WikiText-2 / C4 / BookCorpus, evaluated on each; mirror Loki’s Section 6.3. Include the held-out chunk split.]

[Figure 4. Scatter of per-head relL2 improvement (Loki – CrossCov-U) against ρ_h , with Spearman correlation. One point per (layer, KV head).]

Figure 4: [Per-head advantage vs alignment ratio.]

Table 6: Fraction of the Loki→ J -opt gap closed by each closed-form basis, pre-RoPE, averaged over heads (and over high- ρ_h heads). [Fill.]

Basis	J (closed form)	Empirical score MSE	Recall $r=8$	Gap closed
Loki	x.xx	x.xx	x.xx	0%
SWAN	x.xx	x.xx	x.xx	x.xx%
CrossCov-U	x.xx	x.xx	x.xx	x.xx%
J -opt oracle	x.xx	x.xx	x.xx	100%

Sanity checks. Feeding keys as both sides of the calibration reproduces Loki’s components to $[10^{-x}]$ (Lemma 4.13). The asymmetric $\mathbf{U}/\mathbf{V}$ variant gives recall $[\approx k/S]$ at every r , as predicted by Lemma 4.2.

7.4 Computational efficiency

[Report Loki’s kernel speedup numbers reproduced on our hardware for the selection family (e.g. attention-step time vs sequence length for $r/d = 1/4$, $k/S = 1/8$) and DRAM bytes from Nsight Compute. State explicitly that CrossCov-U and Loki run the same kernels and have identical timing; this subsection exists to show that the accuracy gains come at no runtime cost.]

8 Limitations and Future Work

Surrogate assumptions. The decomposition of Theorem 4.5 closes the fourth moment with a Gaussian assumption and targets the pre-softmax score. We validate both bridges empirically (surrogate ranking agrees with empirical score MSE; score MSE tracks recall, relL2 and perplexity), but neither is a theorem about the end task.

Selection, not compression. CrossCov-U keeps the full KV cache. Section 5.5 explains why the selection basis cannot double as a storage basis; a two-basis design that stores in a key-energy basis and selects with CrossCov-U is the natural extension.

Long-context transfer. Post-RoPE bases average over calibration positions. Whether they, or pre-RoPE bases, transfer to sequences much longer than the calibration length is an open question raised by ShadowKV; our 2×2 experiment is at [length].

Table 7: Basis statistic $\times$ rotary treatment, with fit and inference always matched. [Fill.] The wedge attributable to the statistic is the column difference at fixed row.

Fit/apply	Key-PCA (Σ_k)	Cross-cov ($\mathbf{C}$)
pre-RoPE	recall x.xx / PPL x.xx	recall x.xx / PPL x.xx
post-RoPE	recall x.xx / PPL x.xx	recall x.xx / PPL x.xx

Dynamic bases. $\mathbf{C}$ is a corpus-level statistic. A per-sequence or per-prefix $\mathbf{C}$ would adapt to the prompt at the cost of an online SVD.

9 Conclusion

The basis that sparse attention uses to shrink queries and keys should be chosen for the scores it preserves, not for the keys it reconstructs. We made this precise by decomposing the expected squared score error into an alignment term and a fluctuation term, showed that key-PCA minimizes a degenerate case of the latter and ignores the former, and proposed CrossCov-U, whose basis is the top left singular subspace of the query-key cross-covariance and is surrogate-optimal for the alignment term, with an exactly optimal GQA-pooled form. Changing this one offline statistic, and nothing else, gives [summary of gains] at identical runtime cost.

Reproducibility

Code: [URL]. Calibration scripts, basis artifacts with metadata, and the oracle optimizer are released. [Compute estimate, as in Loki’s Appendix F.]

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A Additional Derivations

A.1 Relation between score error and top- k membership

Let s_j be true scores, $\tilde{s}_j = s_j - \epsilon_j$ approximate scores with $\epsilon_j = \mathbf{q}^\top \mathbf{M} \mathbf{k}_j$, and τ the k -th largest true score. A token j with $s_j \geq \tau$ is dropped only if some token i with $s_i < \tau$ satisfies $\tilde{s}_i > \tilde{s}_j$, which requires $\epsilon_j - \epsilon_i > s_j - s_i \geq s_j - \tau + \tau - s_i$. Hence a wrong swap between j and i needs $|\epsilon_j| + |\epsilon_i| >$ their combined margin to τ . By Markov’s inequality on ϵ_j^2 , the probability that token j ’s error exceeds a margin m_j is at most J/m_j^2 under the calibration distribution, which is how J enters ranking error.

A.2 Frobenius surrogate and the alignment terms: tightness

Equality in Proposition 4.10’s first inequality holds when $\mathbf{M}\mathbf{C}$ is symmetric; in the second when $\mathbf{M}\mathbf{C} \propto \mathbf{M}$ on $\text{range}(\mathbf{M})$. Both hold for symmetric PSD $\mathbf{C}$ in its eigenbasis, which is why Proposition 4.11 is exact there.

A.3 Whitened variant

The exact Eckart–Young statement for the score $\mathbf{q}^\top \mathbf{k}$ under general Σ_q, Σ_k is obtained by whitening: with $\tilde{\mathbf{C}} = \Sigma_k^{-1/2} \mathbf{C} \Sigma_q^{-1/2}$, the best rank- r approximation of $\tilde{\mathbf{C}}$ minimizes $\mathbb{E}[(\mathbf{q}^\top \mathbf{k} - \tilde{s})^2]$ over *all* bilinear rank- r forms, not only shared orthogonal projectors. The resulting key- and query-side maps differ, which brings back the asymmetric full-rank problem of Lemma 4.2 unless the maps are re-orthonormalized. **[Report the whitened variant as an ablation only if its projector is re-orthonormalized; otherwise omit.]**

B Oracle Optimization on the Grassmannian

We minimize $J_{\text{grp}}(\mathbf{U})$ over $\mathbf{U}^\top \mathbf{U} = \mathbf{I}_r$ using the pre-RoPE closed forms of (12). Gradient with respect to $\mathbf{M}$ (treating $\mathbf{M}$ as a free symmetric matrix, then projecting):

$$\nabla_{\mathbf{M}}(\text{tr } \mathbf{M}\mathbf{C})^2 = 2(\text{tr } \mathbf{M}\mathbf{C}) \mathbf{C}^\top, \quad (13)$$

$$\nabla_{\mathbf{M}} \text{tr}(\mathbf{M}\mathbf{C}\mathbf{M}\mathbf{C}) = \mathbf{C}\mathbf{M}\mathbf{C} + \mathbf{C}^\top \mathbf{M}\mathbf{C}^\top, \quad (14)$$

$$\nabla_{\mathbf{M}} \text{tr}(\mathbf{M}\Sigma_q \mathbf{M}\Sigma_k) = \Sigma_q \mathbf{M}\Sigma_k + \Sigma_k \mathbf{M}\Sigma_q. \quad (15)$$

Since $\mathbf{M} = \mathbf{I} - \mathbf{U}\mathbf{U}^\top$, $\nabla_{\mathbf{U}} J = -2\text{sym}(\nabla_{\mathbf{M}} J) \mathbf{U}$. We take the Riemannian gradient $(\mathbf{I} - \mathbf{U}\mathbf{U}^\top) \nabla_{\mathbf{U}} J$, step, and retract by QR. Initialize at the CrossCov-U basis; $d = 128$, $r \leq 64$, **[iterations, learning rate, convergence tolerance]**. Runtime per KV head: **[seconds]**.

C Negative Result: Asymmetric U/V Projection

[Table of recall vs r for keys $\rightarrow \mathbf{U}$, queries $\rightarrow \mathbf{V}$ on Llama-2-7B, showing recall at the k/S floor for pre-RoPE and flat post-RoPE, with the explanation of Lemma 4.2.]

D Complete Results

[Full (r, k) sweep for every model, mirroring Loki’s Appendix B.]